PALAEOMAGNETISM IN THE OSLO RIFT ZONE

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ABSTRACT. Recent palaeomagnetic results (both published and unpublished) from the Oslo region, including data from plutonics, rhomb porphyry lavas, interbasaltic sedimentary horizons and dike rocks, are reviewed. These results suggest that in the southern and western areas of the rift zone there has been substantial magmatism of pre-Kiaman age (i.e. pre ca. 300 m.y. B.P.). In pre-Mesozoic times the areas of main igneous activity appear to have propagated northward. Thus, volcanic activity seems to have swept the Oslo district in the late Carboniferous and/or early Permian while in the northern extremity of the rift (at Brumunddal) outburst of lava flows took place in the latest Permian. Tectonic deformation is seen to post-date the volcanism and is probably of Mesozoic age. At least two pulses of dike intrusions appear to have affected the Oslo region in post-Palaeozoic times, probably in the Jurassic.

1. INTRODUCTION

The original palaeomagnetic study of the Oslo volcanics [14] was for a long time considered as one of the most elaborated palaeomagnetic studies ever undertaken. However, by analysing the data more closely it has become clear that even after alternating field treatment up to several hundred oersteds, within-site magnetizations frequently exhibit meridional distributions. This may imply that one actually is dealing with unresolved multi-component remanences, possibly related to a long magmatic history. These aspects formed the impetus for further palaeomagnetic work in the Oslo region. Of newer published data there are those from 9 diabase dikes in the Kragerø Archipelago [9] and from 3 rhomb porphyry lavas in the

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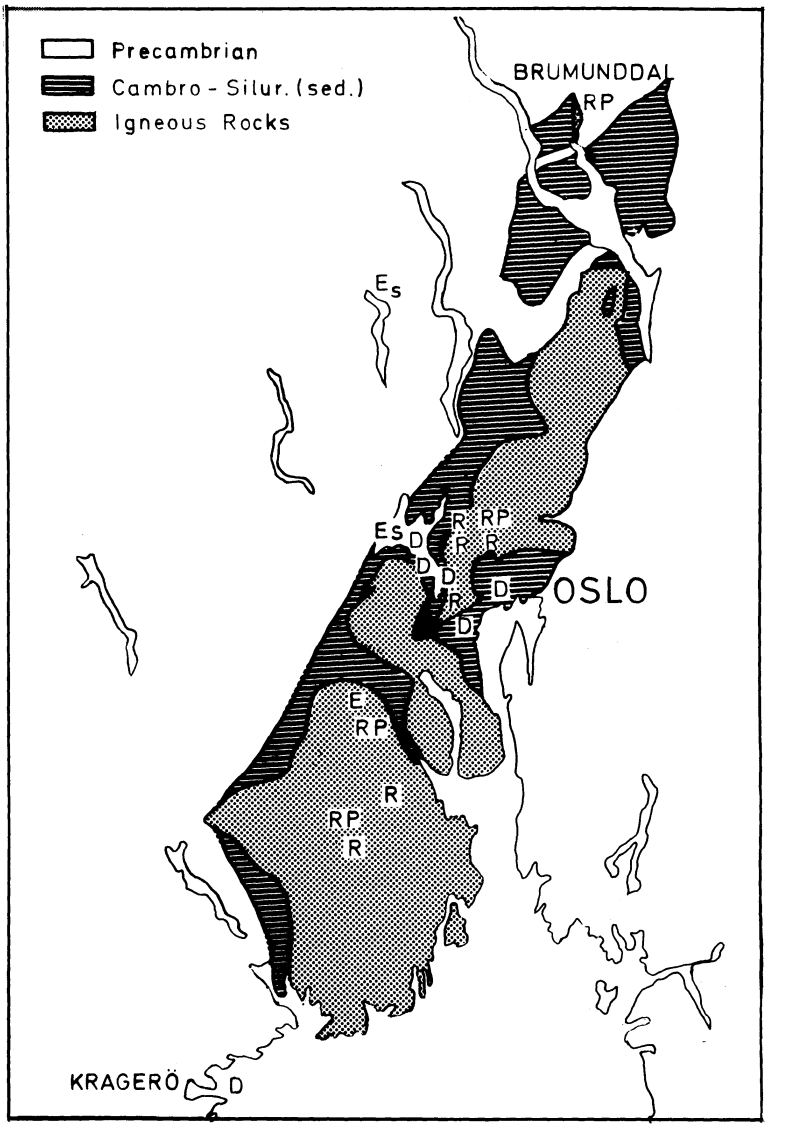


Fig. 1. Geological sketch map of the Oslo rift zone including areal distribution of investigated rocks. R: Red interbasaltic sediments; RP: Rhomb porphyry lavas; E: Ekerite; Es: Essexite; and D: Diabase dikes. In some local areas where more than one site of a particular rock type have been studied these are only shown by a single symbol.

northern extremity of the rift zone [11]. Further data are now available from 23 sites, embracing 3 rhomb porphyry (RP) lavas, 8 interbasaltic sedimentary horizons (baked by overlying RP flows), 1 ekerite (soda granite), 2 essexites and 9 diabase dikes. Distribution of sampling sites is shown in Fig. 1. In the present context only the most important implications of the newer palaeomagnetic data are summarized. Experimental details and further interpretational aspects of the results will be dealt with in a forthcoming paper.

2. RESULTS

As a general rule the magnetization of the plutonic and volcanic rocks, and to some extent also the interbasaltic sediments, is of polyphase structure, consisting of sub-components of opposite polarity and which may cause widely dispersed directions of the resultant magnetization vectors. However, including the results from the Brumunddal lavas [11], a total of five different magnetization axes have now been retrieved. The palaeomagnetic component which is interpreted as the oldest one is termed the A-component (cf. Fig. 2). This magnetization is found in the southern and western areas of the rift zone and constitutes the principal magnetization in the investigated plutonic and volcanic rocks. Fig. 3 demonstrates the magnetization build-up in one of these sites. The A-pole (see Fig. 4) plots in close agreement with the important European Palaeozoic pole position at ca. 20N, 160E and which seems to have been stationary from at least Ordovician to around Middle Carboniferous [8]. The subsequent polar shift into the Permo-Carboniferous pole position (pole 2 of Fig. 4) seems to correlate with the onset of the Kiaman geomagnetic interval (reversed) which is dated to ca. 300 m.y. B.P. [3]. This signifies that at least in the southern and western areas of the rift considerable igneous activity must have occurred prior to 300 m.y. B.P.

On a whole, the investigated interbasaltic sediments form a separate group of palaeomagnetic directions which is termed the B-component (cf. Fig. 2). The angular difference between the A and B magnetizations is nearly 40 degrees and statistically the two mean directions are significantly different. There is some evidence that the two sedimentary sites in the southern districts (cf. Fig. 1) contain the pre-Kiaman axis of magnetization along with a strong Kiaman overprint but this seems not to be the case for the other sedimentary sites which come from the districts around Oslo. After applying tectonic correction the B-pole plots in extremely good agreement with the Permo-Carboniferous pole position (pole 2 of Fig. 4). This suggests that at least the Oslo district was swept by volcanism in late Carboniferous or Permian times. Also, the structural tilt must be post-volcanic, a conclusion which is supported by geological evidence [4]. Unfortunately,

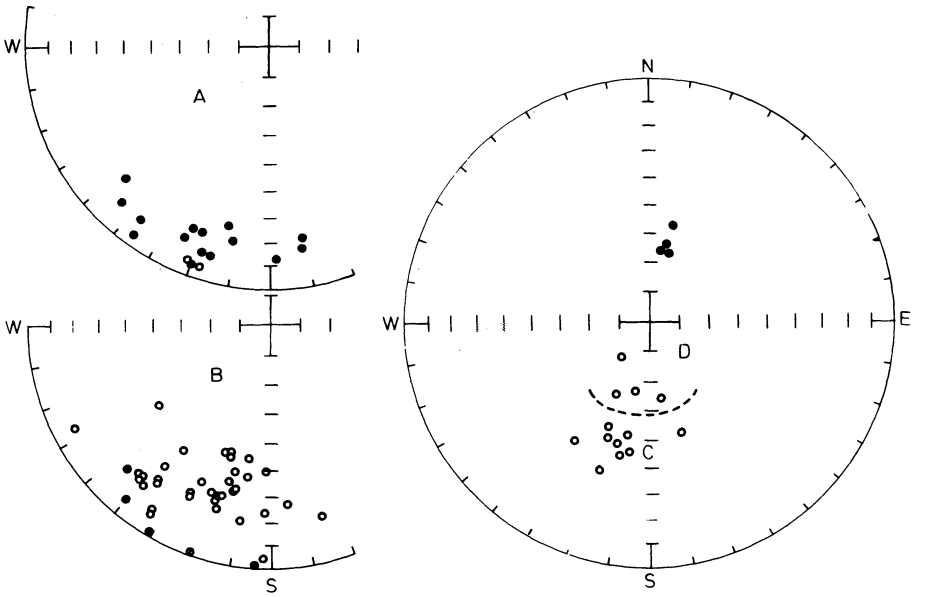


Fig. 2. Palaeomagnetic components from the Oslo rift zone. The A- and B-magnetizations are from plutonic/volcanic rocks and interbasaltic baked sediments respectively. These two components are represented by specimen directions. Magnetizations C and D are from dike rocks in the Oslo and Krageröø districts (cf. Fig. 1). Each dike shows a very good internal consistency of its remanence directions and is believed to constitute a 'point' in time (original). Components C and D are therefore represented by dike mean directions. Note the practically perfect antiparallelism for dike mean directions of the D-component. Data are without tectonic correction. Open (closed) symbols: upward (downward) pointing magnetization vectors. Projection is equal area.

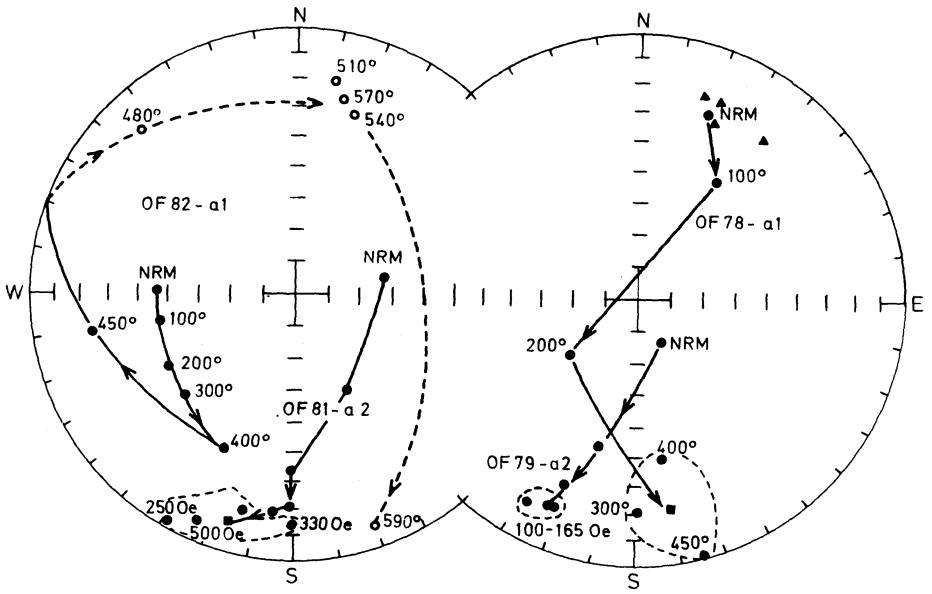


Fig. 3. Directional pattern versus progressive demagnetization for specimens from the ekerite. Note the two-polarity magnetization build-up in specimen OF 82-a1: extended great circle paths through the temperature ranges of NRM-400°C and from 540°-590°C respectively would intersect in a position that agree with stable end-point data from the same site (cf. OF 78-a1, OF 79-a2 and OF 81-a2) and would be closely antiparallel to the normal magnetization directions for the interval 510°-570°C. The triangles represent various vector subtraction estimates for specimen OF 78-a1, defining the erasing vector within the temperature range NRM-300°C. Squares denote the mean high temperature directions (shallow reversed) for individual specimens. Diagram conventions as for Fig. 2.

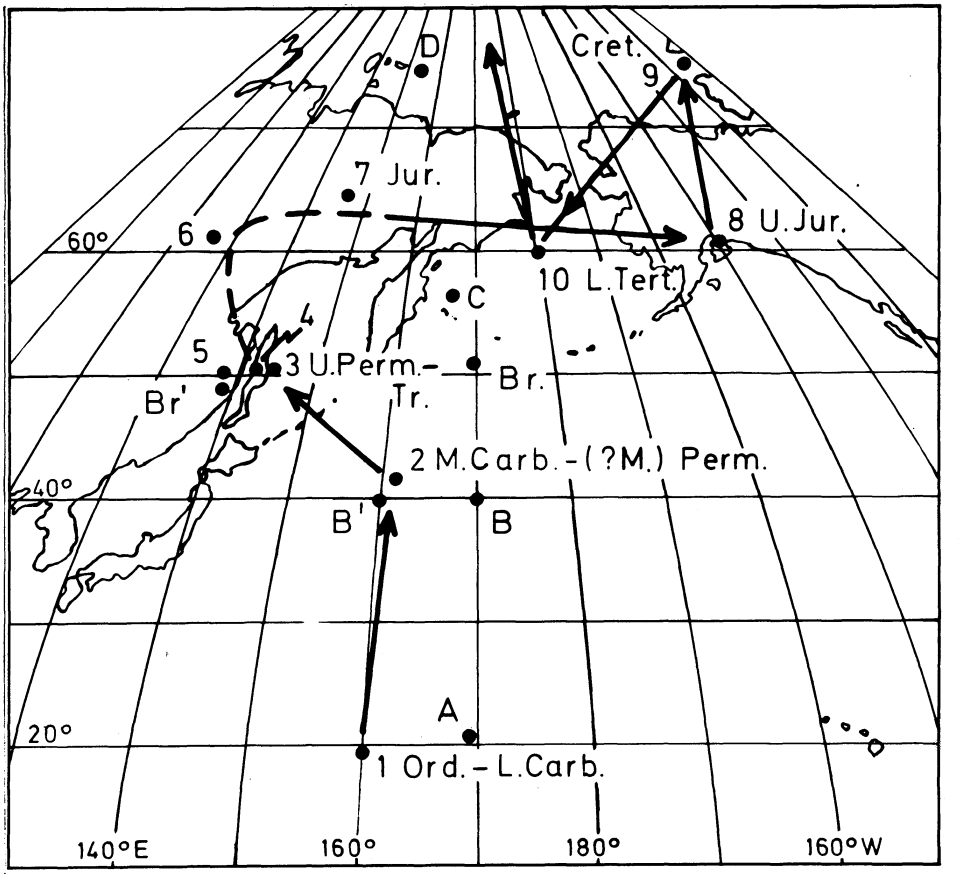


Fig. 4. The Oslo region palaeomagnetic poles seen in conjunction with the suggested European polar wander path for part of Phanerozoic time. Poles 1, 2, 3 and 7 are mean poles from the following sources: 1 [8]; 2, 3 and 7 [5] (other sources give similar results). Individual poles are as follows: 4 [15]; 5 [12]; 6 [13]; 8 [1]; 9 [6]; 10 [10]. A-D refers to the magnetization components of Fig. 2. Br denote the Brumunddal pole [10]. Poles after tectonic correction are shown by dashed symbols (B' and Br'). The Jurassic polar path is very uncertain. With reference to the poles of Fig. 4 an alternative path may be: 6(7)-D-C-8.

at present there are no results from the lavas immediately overlying the investigated sediments, but the available palaeomagnetic evidence from lavas and sediments indicates that the various RP lava units may be of much more local extent than previously suggested [4].

The results from the Brumunddal lava series [11] were rather unexpected in that the Kiaman geomagnetic field (reversed) was only very weakly developed in these rocks while, on the other hand, the Permo-Triassic field axis (both polarities) was very pronounced. The lavas were severly oxidized and in view of a supposedly Lower Permian age for all rhomb porphyries in the Oslo rift zone one was forced to postulate that some unknown process had delayed the oxidation process over a time span of the order of at least 40 m.y. However, by considering the very porous character of the overlying Brumunddal Sandstone (supposed to be a direct continuation of the sedimentary horizons between the lavas) this explanation hardly seems tenable now: a much more straight forward explanation of the data is that the Brumunddal volcanism took place at around the termination of the Kiaman geomagnetic epoch (at ca. 240-230 m.y. B.P.) so that the major oxidation of these rocks occurred in the Permo-Triassic. A late Permian age for the Brumunddal lavas also lines up with the recent suggestion that the Brumunddal Sandstone may be of Triassic age [7]. As seen in Fig. 4 the tectonically corrected Brumunddal pole is in excellent agreement with the late Permian-Triassic palaeomagnetic pole. This implies that the tectonic disturbance, probably connected with the subsidence of the rift, is of post-Permian age.

Several of the dikes collected in the Oslo district are unstably magnetized and reliable palaeomagnetic directions for such rocks have not been obtained. 8 dikes, however, have given very well defined palaeomagnetic results and these are in complete agreement with previously published data from 9 dikes in the Kragerø district (cf. Fig. 1) [9]. The results tend to be grouped along two different axes of magnetization. The combined Oslo and Kragerø results are shown in Fig. 2 as the C and D magnetizations respectively. The C and D poles plot in Mesozoic-Tertiary positions (Fig. 4). A further palaeomagnetic age interpretation of these results is difficult, primarily because of poor knowledge of the Jurassic polar path for Europe. Recent data [2] suggest, however, that pole D may be of Jurassic age and this age may well apply for pole C too. It is therefore tentatively suggested that major dike swarms in the Oslo rift were injected at the time the Central North Sea graben was formed.

Major conclusions:

I Magmatism in the Oslo rift zone initiated prior to 300 m.y. B.P.

II Volcanism covered a time span from pre 300 m.y. B.P. to ~ 240 m.y. B.P.

III In pre-Mesozoic times the igneous centres appear to have propagated northward.

IV Tectonic tilt is post-volcanic and most likely of Mesozoic age.

V Two generation of dikes were injected in Mesozoic times.

REFERENCES

1. E. Briseid and E. Halvorsen, Phys. Earth Planet. Int., 9, 45, 1974.
2. F. Heller, J. Geophys., 42, 475, 1977.
3. M.W. McElhinny, Palaeomagnetism and plate tectonics, Cambridge Univ. Press, Cambridge, 1973.
4. Chr. Oftedahl, Permian rocks and structures of the Oslo region, Norges Geol. Unders., 212, 298, 1960.
5. J.D. Phillips and D. Forsyth, Geol. Soc. Am. Bull., 83, 1579, 1972.
6. S.T. Sandal and E. Halvorsen, Phys. Earth Planet Int., 7, 125, 1973.
7. N. Spjeldnæs, Abstract - Nordisk geol. vintermøte, Oslo, 1972.
8. K.M. Storetvedt, Earth Planet. Sci. Lett., 3, 444, 1967.
9. K.M. Storetvedt, Norsk Geol. Tidsskr., 48, 154, 1968.
10. K.M. Storetvedt, Phys. Earth Planet. Int., 1977 (in press).
11. K.M. Storetvedt and N. Petersen, Z. Geophys., 36, 569, 1970.
12. K.M. Storetvedt and R. Steel, Scott, J. Geol., 1977 (in press).
13. K.M. Storetvedt, C.M. Carmichael, A. Hayatsu and H.C. Palmer, Phys. Earth Planet. Int., 1977 (in press).
14. R.O. Van Everdingen, Skrifter - Det Norske Videnskapsakademi, Mat-Nat. kl., No. 1, Oslo, 1960.
15. J.D.A. Zijderveld, Paleomagnetism of the Esterel rocks, Thesis (unpublished), Utrecht, 1975.